FRE-GY 9743, Assignment 6, Part I

An EUR/USD Forward Collateralised in GBP

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Notation

The assignment's $\mathrm{df}^{[\cdot,\cdot]}$ is a pair, so to keep the derivation unambiguous define

$$D_c^y(T) \;=\; \text{today's value, in currency } y \text{ at spot, of one unit of currency } y \text{ paid at } T, \text{ when the trade is fully collateralised in currency } c.$$

The self-collateralised case is the ordinary OIS curve: $D^{\mathrm{GBP}}_{\mathrm{GBP}} = \mathrm{df}^{\mathrm{SONIA}}$, $D^{\mathrm{EUR}}_{\mathrm{EUR}} = \mathrm{df}^{\mathrm{EuroSTR}}$, $D^{\mathrm{USD}}_{\mathrm{USD}} = \mathrm{df}^{\mathrm{SOFR}}$.

Section 4 maps this onto the assignment's notation, under both readings of the pair. $S(0)$ is EUR/USD spot in the market convention of USD per EUR, and $X_{a\to b}$ is the rate converting one unit of a into b.

1 Q1. The valuation formula

1.1 Framework

Under a CSA that is fully collateralised in currency c with the collateral remunerated at that currency's overnight rate c_t, the collateral account is the numeraire, and the value in currency c of any payoff is

$$V^c(t) = \mathbb{E}_t^{\mathbb{Q}_c}\left[e^{-\int_t^T c_s\,\mathrm{d}s}\,V^c(T)\right]. \tag{1}$$

This is the point of Assignment 4: posting collateral is not settlement, the trade keeps its value, and what the collateral earns decides the discount rate.

Apply (1) to the single cashflow "one unit of currency y at T", whose value at T in currency c is $X_{y\to c}(T)$:

$$\mathbb{E}_t^{\mathbb{Q}_c}\left[e^{-\int_t^T c_s\,\mathrm{d}s}\,X_{y\to c}(T)\right] \;=\; X_{y\to c}(t)\,D_c^y(t,T), \tag{2}$$

which is the definition of D_c^y written as a pricing statement. Note the asymmetry that drives everything below: the *cashflow* currency is y but the *discount* is governed by c, so D_c^y is a genuinely two-index object. Only when $c = y$ does it collapse to a plain OIS curve.

1.2 The forward

Long N EUR of an EUR/USD forward at strike K: at T receive N EUR and pay NK USD. Collateral is GBP, so $c =$ GBP throughout. Taking GBP as the valuation currency and applying (2) to each leg,

$$V^{\mathrm{GBP}}(0) = N\,X_{\mathrm{EUR}\to\mathrm{GBP}}(0)\,D^{\mathrm{EUR}}_{\mathrm{GBP}}(T) \;-\; NK\,X_{\mathrm{USD}\to\mathrm{GBP}}(0)\,D^{\mathrm{USD}}_{\mathrm{GBP}}(T).$$

Dividing by $X_{\mathrm{USD}\to\mathrm{GBP}}(0)$ to express the result in USD, and using $X_{\mathrm{EUR}\to\mathrm{GBP}}/X_{\mathrm{USD}\to\mathrm{GBP}} = X_{\mathrm{EUR}\to\mathrm{USD}} = S$,

$$\boxed{V^{\mathrm{USD}}(0) = N\Big[\, S(0)\, D^{\mathrm{EUR}}_{\mathrm{GBP}}(T) \;-\; K\, D^{\mathrm{USD}}_{\mathrm{GBP}}(T)\Big].} \tag{3}$$

Two observations worth stating explicitly, because they are what the question is probing:

- **The collateral currency enters only through the discount factors, and it enters both legs.** It changes neither cashflow. A GBP CSA does not make this a GBP trade.
- **No GBP curve appears.** $D^{\mathrm{GBP}}_{\mathrm{GBP}}$ is absent from (3) because the trade has no GBP cashflow. Posting GBP collateral means GBP governs *how* the EUR and USD legs are discounted; it does not introduce a GBP payment to discount. This is why $\mathrm{df}^{\mathrm{SONIA}}$ turns out not to be needed in Section 3. That looks surprising until the asymmetry in (2) is taken seriously.

2 Q2. The fair forward strike

The fair strike is the K that makes (3) vanish:

$$\boxed{K^* = S(0)\,\frac{D^{\mathrm{EUR}}_{\mathrm{GBP}}(T)}{D^{\mathrm{USD}}_{\mathrm{GBP}}(T)}.} \tag{4}$$

Substituting (4) back into (3) gives the form actually used to mark an existing trade:

$$V^{\mathrm{USD}}(0) = N\, D^{\mathrm{USD}}_{\mathrm{GBP}}(T)\,\big(K^* - K\big). \tag{5}$$

The split in (5) matters for Section 3: the *strike* depends on the *ratio* of the two GBP-collateralised curves, while the *value* additionally needs one of them on its own.

3 Q3. Are the five discount factors sufficient?

3.1 What is needed, and what is missing

Equations (3) and (4) require exactly two objects: $D^{\mathrm{EUR}}_{\mathrm{GBP}}$ and $D^{\mathrm{USD}}_{\mathrm{GBP}}$. Of these, $D^{\mathrm{USD}}_{\mathrm{GBP}}$ is supplied directly. **Strictly, $D^{\mathrm{EUR}}_{\mathrm{GBP}}$ is not in the list**. The EUR curve under GBP collateral is precisely the one that is absent.

So the literal answer is: not sufficient as a matter of pure bookkeeping. The useful answer is that the missing curve is *constructible* from the rest, provided one additional assumption is stated rather than smuggled in.

3.2 The assumption that closes the gap

The relevant assumption is that a fully collateralised FX forward has the same fair rate regardless of which currency the CSA specifies. Equivalently: for any pair (a, b) and any collateral currency c,

$$F_{a\to b}(T) = X_{a\to b}(0)\,\frac{D^a_c(T)}{D^b_c(T)} \quad \text{is independent of } c. \tag{6}$$

In basis language this is triangular consistency of the cross-currency bases: writing $D^y_c = D^y_y \exp\!\big(-\int b_{c,y}\big)$, (6) is the statement $b_{\mathrm{GBP},\mathrm{EUR}} = b_{\mathrm{GBP},\mathrm{USD}} - b_{\mathrm{EUR},\mathrm{USD}}$. It is the assumption every curve desk makes

when building a non-USD-collateral basis, because USD is the hub currency where cross-currency swap liquidity actually sits.

Applying (6) to the EUR/USD pair with $c = \text{GBP}$ and $c = \text{EUR}$:

$$\frac{D^{\text{EUR}}_{\text{GBP}}}{D^{\text{USD}}_{\text{GBP}}} = \frac{D^{\text{EUR}}_{\text{EUR}}}{D^{\text{USD}}_{\text{EUR}}} \qquad \Longrightarrow \qquad D^{\text{EUR}}_{\text{GBP}} = D^{\text{USD}}_{\text{GBP}}\, \frac{D^{\text{EUR}}_{\text{EUR}}}{D^{\text{USD}}_{\text{EUR}}},$$

and every term on the right is supplied. **So the answer is yes: sufficient, conditional on** (6)**.**

3.3 What the formulas collapse to

Substituting into (4), the GBP factor cancels:

$$K^* = S(0)\, \frac{D^{\text{EUR}}_{\text{EUR}}(T)}{D^{\text{USD}}_{\text{EUR}}(T)} = S(0)\, \frac{\text{df}^{\text{EuroSTR}}(T)}{\text{df}^{[\text{EUR,USD}]}(T)}. \tag{7}$$

This is the whole content of the question in one line. **The fair strike does not depend on the collateral currency at all.** That is not a coincidence or a lucky cancellation. It is (6) restated, since that assumption says exactly that the forward rate is collateral-invariant. A GBP CSA buys you nothing and costs you nothing on a trade struck at the market.

The value of an *off-market* forward is a different matter. From (5),

$$V^{\text{USD}}(0) = N\, \text{df}^{[\text{GBP,USD}]}(T)\, (K^* - K), \tag{8}$$

and here the GBP collateralisation genuinely bites: the payoff $(K^* - K)$ is discounted on the GBP-collateralised USD curve, not on SOFR. Two desks holding the same trade under different CSAs agree on the fair strike and disagree on the mark.

3.4 Which of the five are actually used

Curve	Role	Needed?
$\text{df}^{\text{EuroSTR}}$	numerator of the strike (7)	yes
$\text{df}^{[\text{EUR,USD}]}$	denominator of the strike (7)	yes
$\text{df}^{[\text{GBP,USD}]}$	discounting of the mark (8)	yes
df^{SOFR}	USD under *USD* collateral; wrong CSA here	no
df^{SONIA}	GBP cashflows; the trade has none	no

Two of the five are redundant for this trade, and the two that are redundant are the ones a careless answer reaches for first: SONIA because the collateral is GBP, SOFR because the trade settles in USD. Neither reason survives (2). SOFR would be needed if the CSA were in USD; SONIA only if the trade paid GBP.

3.5 Mapping to the assignment's notation

The pair $\text{df}^{[\cdot,\cdot]}$ admits two readings, and the structural answer is the same either way; only the arithmetic route changes.

Reading A, $\text{df}^{[c,y]}$ **is cashflow** y **under collateral** c **(used above).**
$\text{df}^{[\text{GBP,USD}]} = D^{\text{USD}}_{\text{GBP}}$ is given directly, and $D^{\text{EUR}}_{\text{GBP}}$ is built as above. SONIA and SOFR are unused.

Reading B, $\mathrm{df}^{[y,c]}$ is cashflow y under collateral c.

Now $\mathrm{df}^{[\mathrm{EUR,USD}]} = D^{\mathrm{EUR}}_{\mathrm{USD}}$ and $\mathrm{df}^{[\mathrm{GBP,USD}]} = D^{\mathrm{GBP}}_{\mathrm{USD}}$. Applying (6) to EUR/USD gives the strike straight off USD-collateral curves, $K^* = S(0)\,\mathrm{df}^{[\mathrm{EUR,USD}]}/\mathrm{df}^{\mathrm{SOFR}}$, and applying it to the USD/GBP pair gives the discounting curve $D^{\mathrm{USD}}_{\mathrm{GBP}} = \mathrm{df}^{\mathrm{SONIA}}\,\mathrm{df}^{\mathrm{SOFR}}/\mathrm{df}^{[\mathrm{GBP,USD}]}$. Here four of the five are used and EuroSTR is the redundant one.

Under either reading: sufficient given (6), the strike is collateral-independent, and the GBP CSA shows up only in the discounting of the mark.

3.6 When the assumption fails

Equation (6) is a modelling choice, not a theorem, and it is worth being precise about where it breaks:

1. **A directly quoted EUR/GBP basis.** The triangulated $D^{\mathrm{EUR}}_{\mathrm{GBP}}$ is only as good as the claim that EUR/GBP basis equals GBP/USD minus EUR/USD. EUR/GBP cross-currency swaps do trade, and when the quoted basis differs from the triangulated one, the constructed curve is simply wrong. If that market is liquid at the relevant tenor, build $D^{\mathrm{EUR}}_{\mathrm{GBP}}$ from it directly and treat the identity as a check rather than as the source.

2. **Collateral optionality.** A multi-currency CSA that lets the poster choose which currency to deliver makes the discount curve an option on the cheapest-to-deliver basis, not a curve at all. Then no set of single-currency discount factors is sufficient, however large. The question specifies GBP collateral, which avoids this, but it is the first thing to confirm in the actual CSA, because the phrase "collateralised in GBP" is often shorthand for "GBP is one of the eligible currencies".

3. **Collateral convexity.** (6) holds exactly under deterministic rates and FX. With both stochastic, correlation between the collateral rate and the FX rate puts a convexity correction between the collateral-adjusted curves. It is second-order and routinely ignored for short tenors, which is the honest reason the identity is used rather than any claim that it is exact.